How Do Kernel Size and Pooling Affect Feature Learning in CNNs? A Comparative Visual Analysis

GitHub: https://github.com/reecha-24160377/cnn-kernel-pooling-analysis

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Abstract

Convolutional Neural Networks (CNNs) are widely used in image classification due to their ability to automatically learn feature representations from data. This study investigates the impact of two important architectural components of CNNs: kernel size and pooling.

Six CNN models were developed using different kernel sizes (3×3, 5×5, and 7×7), both with and without pooling layers, and evaluated on the MNIST dataset. The models were compared using training and validation accuracy, loss curves, and feature map visualisations.

The results show that smaller kernels (3×3) combined with pooling achieve the best performance and generalisation. Larger kernels tend to lose fine-grained details, while models without pooling are more prone to overfitting. This study provides practical insights into designing effective CNN architectures.

1. Introduction

Machine learning plays a significant role in solving complex problems across various domains such as healthcare, finance, and computer vision. Among different machine learning techniques, Convolutional Neural Networks (CNNs) are particularly effective for image-related tasks due to their ability to automatically learn features from raw data.

CNNs operate by applying convolutional filters to images to detect patterns such as edges, textures, and shapes. These features are learned hierarchically, where early layers capture simple patterns and deeper layers learn more complex representations.

Two important design choices in CNN architecture are kernel size and pooling operations. Kernel size determines how much spatial information is captured, while pooling reduces spatial dimensions and improves generalisation.

2. Background Theory

Convolutional Neural Networks consist of multiple layers, including convolutional layers, pooling layers, and fully connected layers. The convolutional layer plays a key role in feature extraction by applying filters to input images to detect patterns such as edges, textures, and shapes.

Kernel Size

Kernel size refers to the dimensions of the filters used in the convolution process, commonly 3×3, 5×5, and 7×7.

- **Smaller kernels (3×3):**
  Capture fine-grained spatial details such as edges and textures while maintaining computational efficiency.
- **Medium kernels (5×5):**
  Capture broader patterns but may lose some fine detail compared to smaller kernels.

- **Larger kernels (7×7):**
  Capture more global features but often reduce local detail and increase computational complexity.

Pooling

Pooling is used to reduce the spatial dimensions of feature maps, making the model more efficient. The most common type is max pooling, which selects the maximum value within a region.

The advantages of pooling include:

- Reducing computational cost
- Increasing translation invariance
- Reducing overfitting

However, removing pooling layers can lead to unstable learning and increased overfitting.

3. Dataset and Experimental Setup

The MNIST dataset was used for this experiment. It consists of 70,000 grayscale images of handwritten digits, each with a resolution of 28×28 pixels.

To analyse the effect of kernel size and pooling, six CNN models were implemented:

- 3×3 kernel with pooling
- 3×3 kernel without pooling
- 5×5 kernel with pooling
- 5×5 kernel without pooling
- 7×7 kernel with pooling
- 7×7 kernel without pooling

All models were trained using identical parameters:

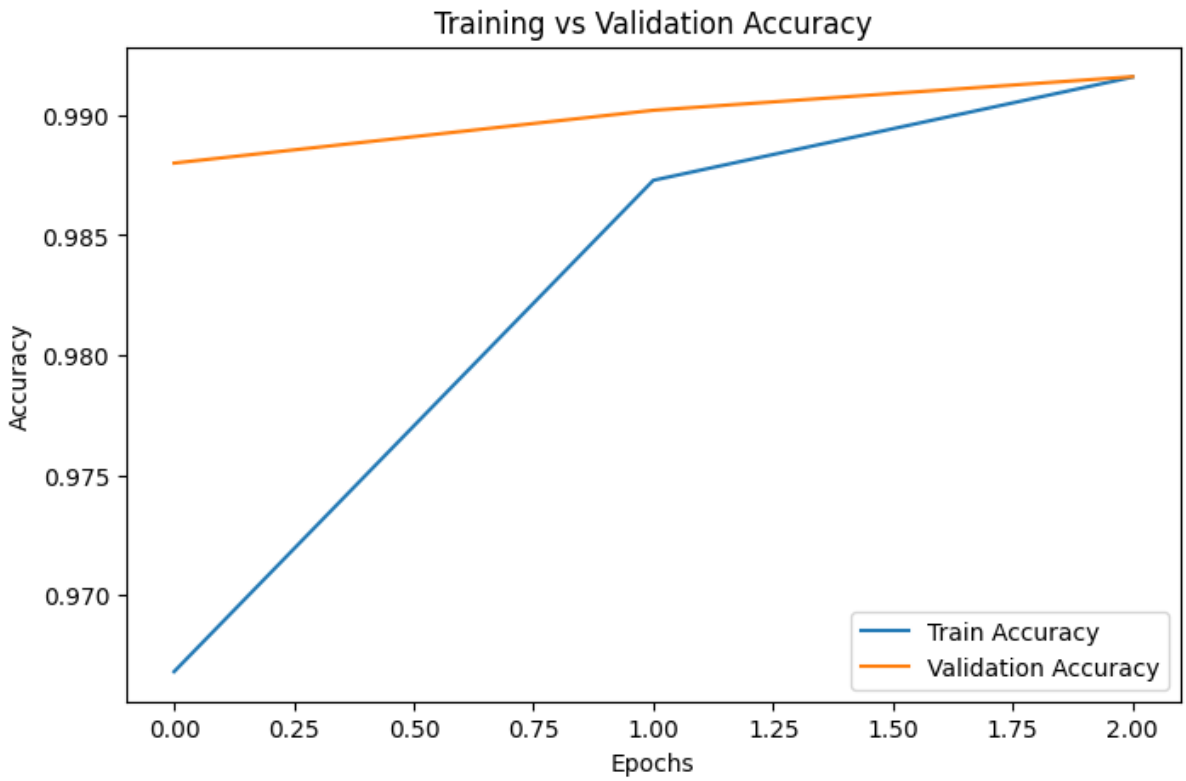
- Adam optimiser
- Sparse categorical cross-entropy loss
- 3–5 epochs

This ensures that differences in performance are due to kernel size and pooling rather than training conditions.

4. Results and Visualisation

Training and Validation Accuracy

Figure 1 shows the training and validation accuracy of a CNN model across training epochs. The model demonstrates steady improvement in both training and validation accuracy, indicating effective learning and good convergence. The small gap between training and validation accuracy suggests that the model generalises well without significant overfitting.



Validation Accuracy Comparison

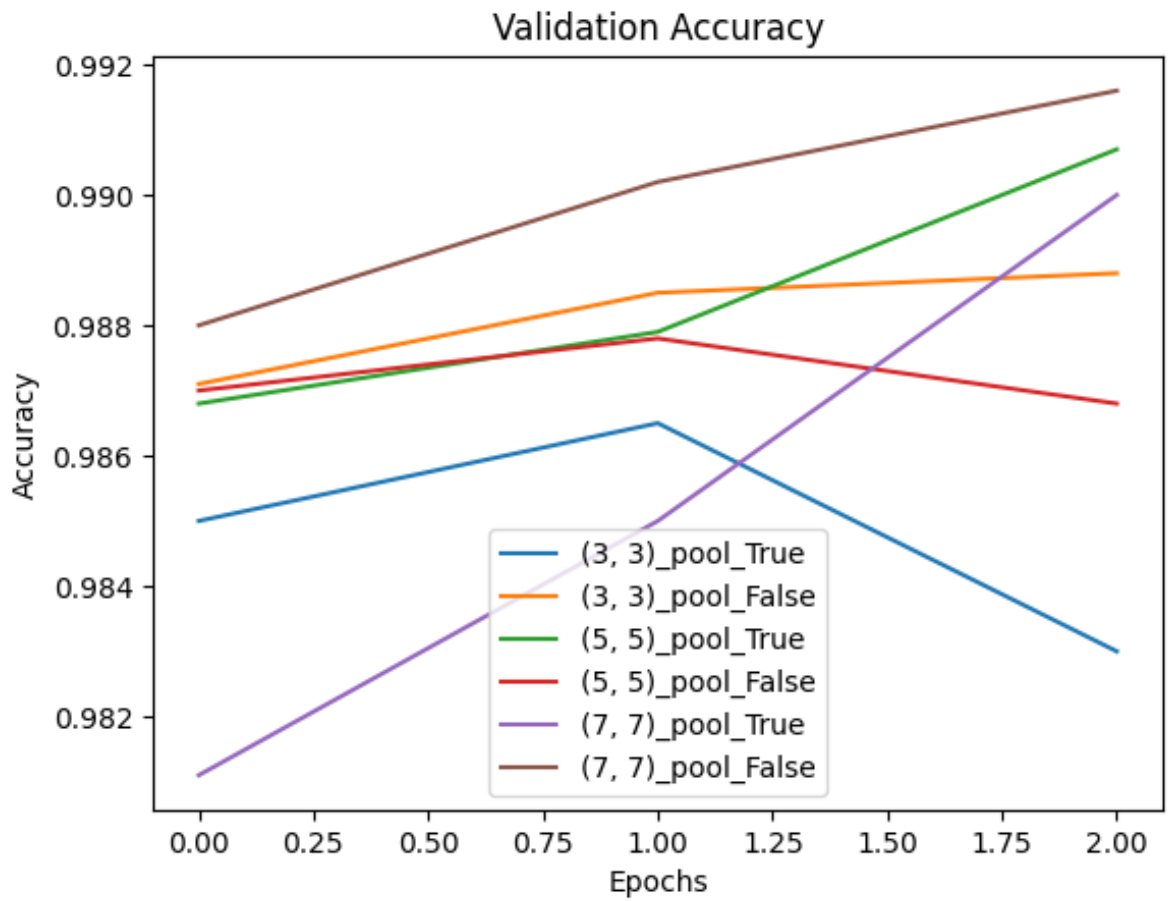


Figure 2 presents a comparison of validation accuracy across all six model configurations. The results show that models with pooling generally achieve more stable and consistent performance. While some configurations such as larger kernels may achieve slightly higher accuracy at certain epochs, smaller kernels (3×3) provide more consistent and reliable performance across training. Models without pooling show less stable behaviour, indicating potential overfitting.

Loss and Validation Loss Analysis

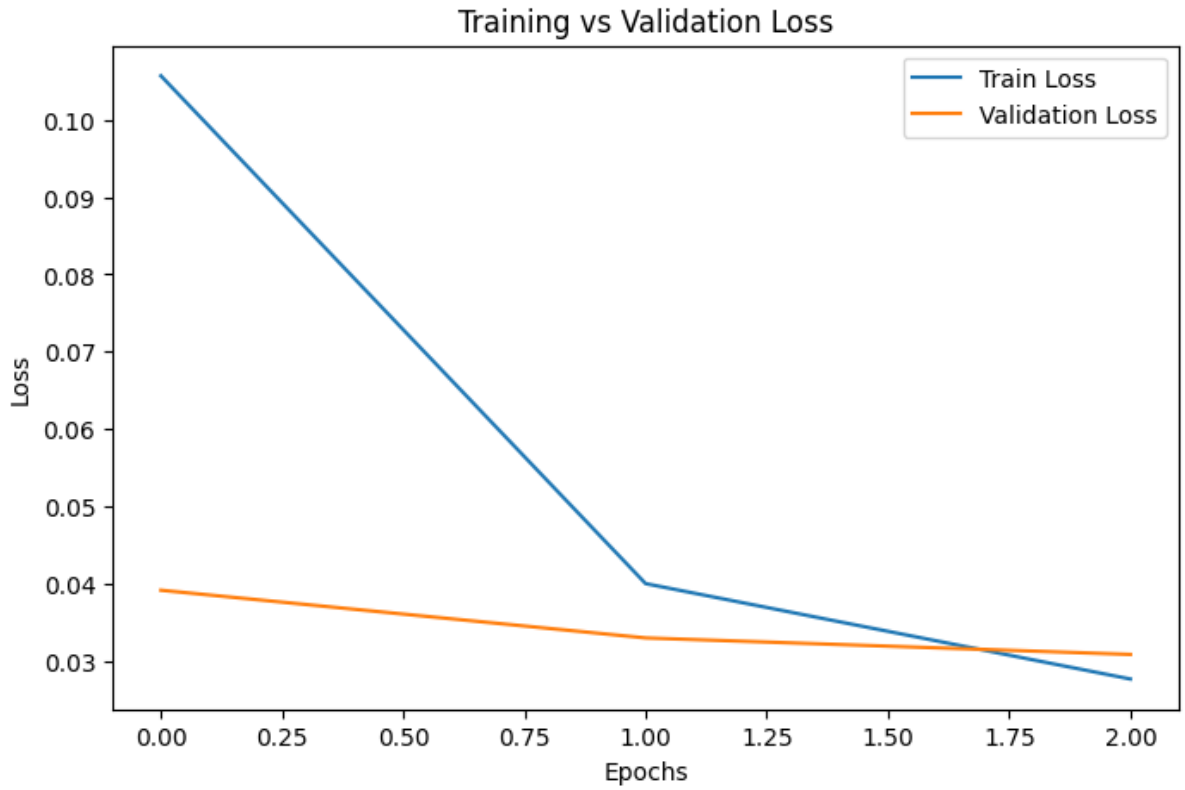


Figure 3 illustrates the training and validation loss curves. Both curves show a decreasing trend, indicating effective learning. The validation loss decreases smoothly, suggesting good generalisation. The small gap between training and validation loss indicates that the model is not significantly overfitting. Overall, the loss curves confirm stable and efficient training behaviour.

Feature Map Visualisation

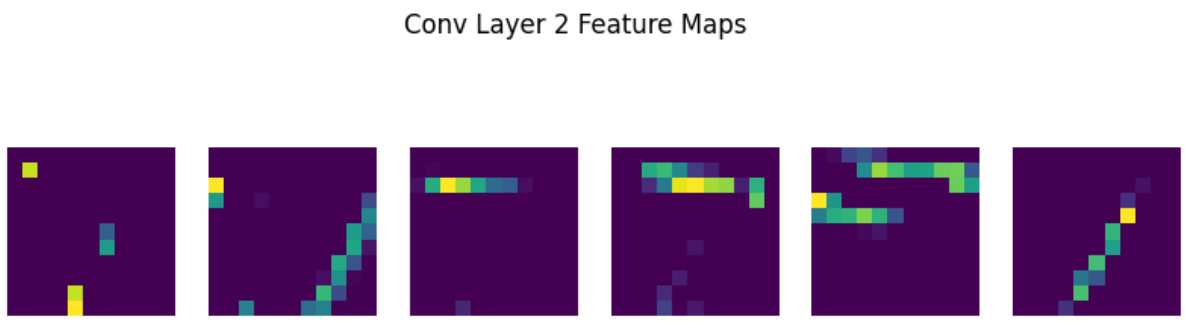


Figure 4 shows the feature maps generated by different convolutional layers. The feature maps from early layers capture simple patterns such as edges and strokes, while deeper layers represent more abstract features. Models using smaller kernels produce clearer and sharper feature maps, indicating better preservation of fine details. In contrast, larger kernels produce smoother and less detailed feature maps, reflecting a focus on global patterns rather than local features.

Effect of Pooling

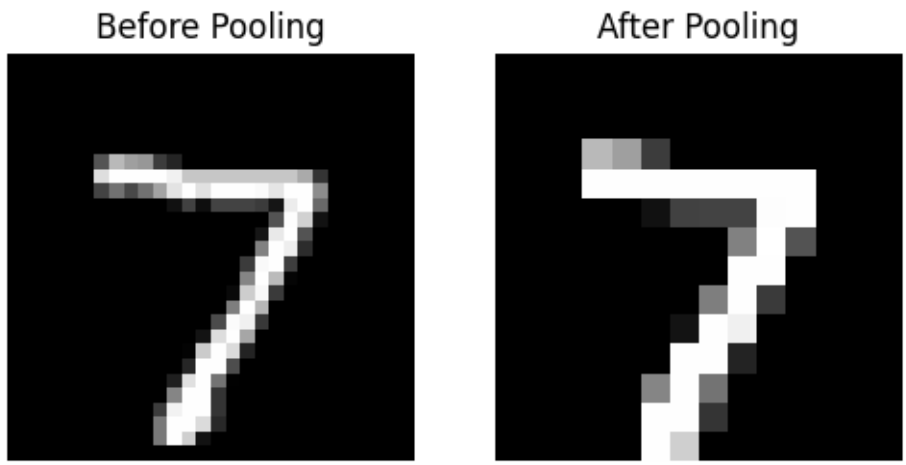


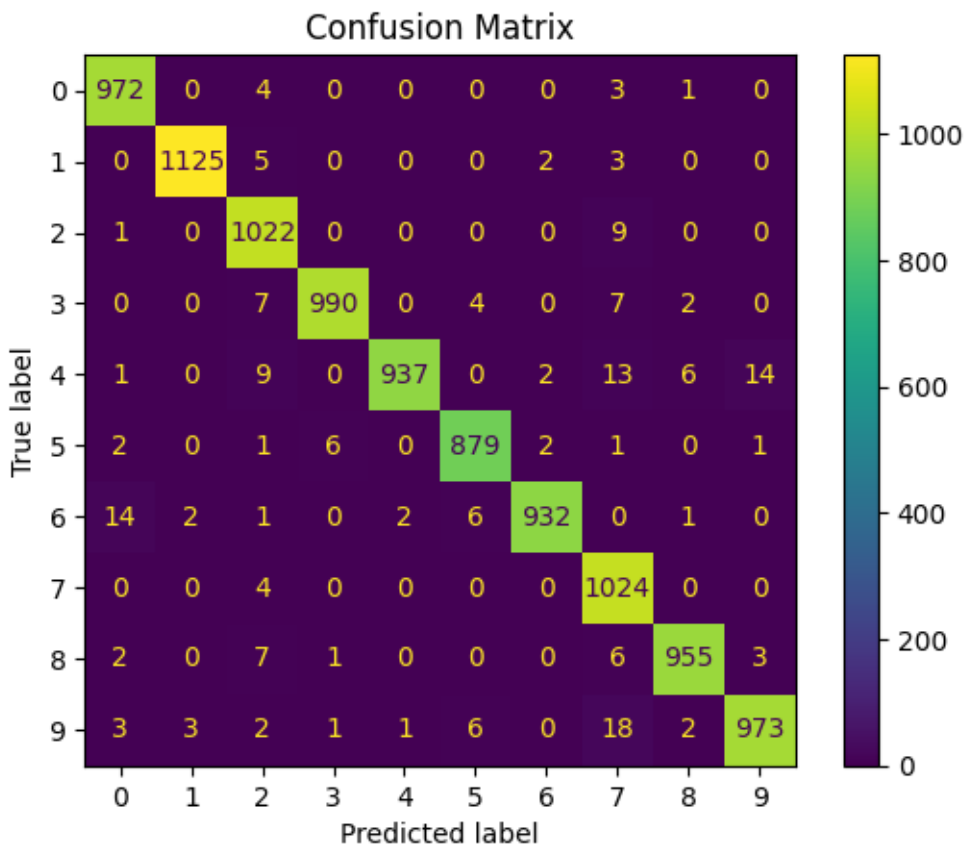
Figure 5 demonstrates the effect of pooling on feature maps. Before pooling, the feature maps contain detailed but noisy information. After pooling, the feature maps become simpler and more focused, highlighting the most important features. This shows that pooling helps reduce noise, improve generalisation, and make the model more efficient.

Confusion Matrix Analysis

Figure 6 shows the confusion matrix for the CNN model. The confusion matrix compares the predicted labels with the actual labels and provides a detailed view of classification performance.

Most of the values are concentrated along the diagonal, indicating that the model correctly classifies the majority of digits. This demonstrates strong overall performance. However, some misclassifications can be observed between visually similar digits, such as 3 and 5 or 4 and 9, highlighting the model's limitations in distinguishing similar patterns.

Overall, the confusion matrix provides deeper insight into model performance beyond accuracy and confirms that the model achieves reliable classification results.



6. Discussion

The findings highlight a trade-off between feature detail and computational efficiency.

Smaller kernels perform better because they preserve fine details and allow efficient feature extraction. Larger kernels increase computational cost and reduce precision.

Pooling plays a crucial role in improving generalisation by reducing overfitting and focusing on important features.

Overall, smaller kernels combined with pooling provide the best balance between performance and efficiency.

7. Practical Takeaways

- 3×3 kernels are the most effective for CNNs
- Pooling improves generalisation and reduces overfitting
- Larger kernels increase complexity without significant benefits
- Feature visualisation helps interpret model behaviour
- Model design should balance accuracy and efficiency

8. Ethical Considerations

Machine learning models may inherit biases present in training data. Although MNIST is simple, real-world datasets may introduce fairness issues.

Additionally, more complex models require higher computational resources, which increases energy consumption and environmental impact.

Responsible model design should consider both fairness and efficiency.

9. Accessibility Considerations

Steps taken to improve accessibility include:

- Clear structure and headings
- Simple and understandable language
- Well-labelled graphs and visualisations
- Readable fonts and layout

These ensure the tutorial is accessible to a wide audience.

10. Conclusion

This study examined the impact of kernel size and pooling on feature learning in CNNs.

The results demonstrate that 3×3 kernels combined with pooling achieve the best performance, as they effectively capture fine details while maintaining generalisation. Larger kernels introduce unnecessary complexity, and removing pooling leads to overfitting.

Overall, kernel size and pooling are critical design choices that significantly influence CNN performance.

References

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). *Deep learning*. Nature, 521(7553), 436–444.
- TensorFlow Documentation: https://www.tensorflow.org/
- MNIST Dataset: http://yann.lecun.com/exdb/mnist/